

H¹-in- ρ Bound for the Centroid of the Torsion Level Family (Route δ , Plan 2, Building Block CENTROIDBOUND)

Abstract

This note is the building block CENTROIDBOUND of the Plan 2 proof of sharp quantitative Faber–Krahn stability via Route δ (no graph, no selection, no Schauder). For a near-spherical torsion domain $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and small Saint–Venant deficit $\delta := \delta_T(\Omega)$, let $E_\rho := \{u > t(\rho)\}$ be the volume-radius parametrised superlevels of the torsion function u and let

$$\bar{z}_\rho := \frac{1}{|E_\rho|} \int_{E_\rho} x \, dx \in \mathbb{R}^n$$

be the *centroid* of E_ρ . We prove that $\rho \mapsto \bar{z}_\rho$ is Lipschitz on $[\rho_*, 1]$ and satisfies the integrated H¹-in- ρ bound

$$\int_{\rho_*}^{\rho_\delta} |\bar{z}'_\rho|^2 (-t'(\rho)) \, d\rho \leq C(n, R, \rho_*) \delta_T(\Omega).$$

The proof uses the centroid kinematic identity (a bulk-coarea computation that has *no* homothety term), the unconditional Cauchy–Schwarz of the velocity defect controlled by $D_H + D_I$, the Figalli–Maggi–Pratelli volume Fraenkel estimate, the Fusco–Julin L^1 oscillation, the divergence theorem for finite-perimeter sets, and the integrated Talenti gap of [9]. No graph parametrisation, no selection principle, no pointwise quantitative isoperimetric form is invoked.

1 Statement and main result

Fix $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$, and $\kappa > 0$. Let $\Omega \subset B_R \subset \mathbb{R}^n$ be a bounded open set with $|\Omega| = \omega_n := |B_1|$ and Saint–Venant deficit $\delta := \delta_T(\Omega) \leq \delta_0(n, R, \rho_*)$, where $\delta_T(\Omega) := 1 - T(\Omega)/T(B_1)$ and $T(\cdot)$ denotes the Saint–Venant torsional rigidity. Let $u \in W_0^{1,2}(\Omega) \cap W_{\text{loc}}^{2,p}(\Omega)$ ($p > n$) be the variational torsion function of Ω , $t : [\rho_*, 1] \rightarrow (0, \|u\|_\infty)$ the volume-radius parametrisation defined by $|\{u > t(\rho)\}| = \omega_n \rho^n$, $E_\rho := \{u > t(\rho)\}$, and

$$V_\rho(x) := \frac{-t'(\rho)}{|\nabla u(x)|} \quad \text{on } \partial^* E_\rho \cap \{|\nabla u| > 0\}.$$

The set $\{|\nabla u| = 0\} \cap \partial^* E_\rho$ has $\mathcal{H}^{n-1}$ -measure zero on G_{CL} (Sard plus regularity, cf. [9]), so V_ρ is $\mathcal{H}^{n-1}$ -a.e. defined on $\partial^* E_\rho$ at good levels. Set $\rho_\delta := 1 - \kappa\sqrt{\delta}$.

Definition 1.1 (Centroid field). For $\rho \in [\rho_*, 1]$ define the *centroid* of E_ρ by

$$\bar{z}_\rho := \frac{1}{|E_\rho|} \int_{E_\rho} x \, dx \in \mathbb{R}^n. \tag{1}$$

This is the bulk barycentre; it is *not* obtained by Fraenkel minimisation and inherits no implicit-function-theorem regularity.

Theorem 1.2 (H^1 -in- ρ centroid bound). *There exist $\delta_0 = \delta_0(n, R, \rho_*) > 0$ and $C_{\text{cent}} = C_{\text{cent}}(n, R, \rho_*) > 0$ such that for every bounded open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta \leq \delta_0$, the centroid field $\rho \mapsto \bar{z}_\rho$ is Lipschitz on $[\rho_*, 1]$ with explicit constant*

$$L_{\text{cent}} := \frac{2nR}{\rho_*^n}, \quad (2)$$

in particular absolutely continuous, and

$$\int_{\rho_*}^{\rho_\delta} |\bar{z}'_\rho|^2 (-t'(\rho)) d\rho \leq C_{\text{cent}} \delta_T(\Omega). \quad (\text{F})$$

Remark 1.3 (Compatibility with the Fraenkel centre). By Cicalese–Leonardi [3, Prop. 2.3] and Acerbi–Fusco–Morini [1, Lem. 2.6], in the smallness regime $|\bar{z}_\rho - z_\rho^{\text{Fr}}| \leq C \mathcal{A}(E_\rho) \leq C' \sqrt{\delta_T(\Omega)}$. The centroid is therefore interchangeable with z_ρ^{Fr} in downstream uses at cost $O(\sqrt{\delta_T})$, absorbed into constants exactly as in Wave 2 C §C1.

2 Setup, measurability, Lipschitz-in- ρ

2.1 Good levels and the CL regime

The exact level-set identity gives the integrated bound

$$\int_{\rho_*}^1 (D_H(t(\rho)) + D_I(t(\rho))) d\mu(\rho) \leq C(n, \rho_*) \delta_T(\Omega), \quad d\mu = (-t') d\rho. \quad (3)$$

It does *not* imply pointwise smallness of $D_I(t(\rho))$ for every ρ . We therefore use the following good-level split. Let $\eta_{\text{CL}}(n, \rho_*) > 0$ be small enough that

$$D_I(t(\rho)) \leq \eta_{\text{CL}} \implies \mathcal{A}(E_\rho) \leq \varepsilon_0(n)$$

by the quantitative isoperimetric inequality. Define

$$G_{\text{CL}} := \{\rho \in [\rho_*, 1] : D_I(t(\rho)) \leq \eta_{\text{CL}}\}, \quad B_{\text{CL}} := [\rho_*, 1] \setminus G_{\text{CL}}.$$

Then $\mu(B_{\text{CL}}) \leq C\delta_T(\Omega)$ by (3). (Chebyshev: on B_{CL} one has $D_I(t(\rho)) > \eta_{\text{CL}}$, so $\eta_{\text{CL}} \mu(B_{\text{CL}}) \leq \int_{B_{\text{CL}}} D_I d\mu \leq \int_{\rho_*}^1 (D_H + D_I) d\mu \leq C(n, \rho_*) \delta_T(\Omega)$, whence $\mu(B_{\text{CL}}) \leq C'\delta_T(\Omega)$.) On G_{CL} , Cicalese–Leonardi gives a Fraenkel centre z_ρ^{Fr} with

$$B(z_\rho^{\text{Fr}}, \rho/2) \subset E_\rho \subset B(z_\rho^{\text{Fr}}, 2\rho). \quad (4)$$

Combined with the centroid–Fraenkel proximity, after decreasing δ_0 if needed,

$$B(\bar{z}_\rho, \rho/4) \subset E_\rho \subset B(\bar{z}_\rho, 3\rho) \quad \text{for a.e. } \rho \in G_{\text{CL}}.$$

In particular $|x - \bar{z}_\rho| \leq 3$ on $\partial^* E_\rho$ and $\bar{z}_\rho \in E_\rho^\circ$ on the good levels. On B_{CL} we use only the global Lipschitz bound of Lemma 2.1.

2.2 Borel measurability and Lipschitz

Lemma 2.1. $\rho \mapsto \bar{z}_\rho$ is Borel and Lipschitz on $[\rho_*, 1]$ with constant $L_{\text{cent}} = 2nR/\rho_*^n$.

Proof. **Borel.** Since t is decreasing, $\rho \mapsto E_\rho$ is monotone nested; dominated convergence gives $\rho \mapsto \int_{E_\rho} x_i dx$ continuous, hence Borel.

Lipschitz. For $\rho_* \leq \rho \leq \rho' \leq 1$, nesting gives

$$|E_\rho \Delta E_{\rho'}| = |E_{\rho'}| - |E_\rho| = \omega_n((\rho')^n - \rho^n) \leq n\omega_n|\rho - \rho'|.$$

Since $E_\rho, E_{\rho'} \subset B_R$, $|\int x_i - \int x_i| \leq Rn\omega_n|\rho - \rho'|$. Combined with $|E_\rho| \geq \omega_n\rho_*^n$, $|\bar{z}_\rho - \bar{z}_{\rho'}| \leq (2nR/\rho_*^n)|\rho - \rho'|$. $\square$

3 The centroid kinematic identity

Lemma 3.1 (Centroid kinematic formula). *For a.e. $\rho \in [\rho_*, 1]$,*

$$\bar{z}'_\rho = \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} (x - \bar{z}_\rho) V_\rho d\mathcal{H}^{n-1}. \quad (5)$$

Proof. By the coarea identity [9, §3] and Sobolev–Sard [4, App. A, Thm. A.1], for every Lipschitz $f : B_R \rightarrow \mathbb{R}$ and a.e. $\rho \in [\rho_*, 1]$, $\frac{d}{d\rho} \int_{E_\rho} f dx = \int_{\partial^* E_\rho} f V_\rho d\mathcal{H}^{n-1}$. By the volume normalisation $|E_\rho| = \omega_n \rho^n$ we have $\frac{d}{d\rho} |E_\rho| = n\omega_n \rho^{n-1} = P(B_\rho)$, and combining with coarea ($f \equiv 1$) gives $\int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = \frac{d}{d\rho} |E_\rho| = P(B_\rho)$. Taking $f(x) = x_i$ in the coarea identity and applying the product rule to $\bar{z}_{\rho,i} = |E_\rho|^{-1} \int_{E_\rho} x_i dx$:

$$\bar{z}'_{\rho,i} = \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} x_i V_\rho d\mathcal{H}^{n-1} - \frac{1}{|E_\rho|^2} \left(\frac{d}{d\rho} |E_\rho| \right) \int_{E_\rho} x_i dx.$$

Substituting the coarea form $\frac{d}{d\rho} |E_\rho| = \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1}$ into the second term yields

$$\bar{z}'_{\rho,i} = \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} (x_i - \bar{z}_{\rho,i}) V_\rho d\mathcal{H}^{n-1},$$

the displayed identity. The equality $\frac{d}{d\rho} |E_\rho| = P(B_\rho)$ is here a *consequence* of the volume normalisation, not a perimeter identity for E_ρ . $\square$

Remark 3.2 (No homothety term — the structural mechanism). Identity (5) is the first vector moment of V_ρ relative to the centroid. *No homothetic vector field $H_{z,\rho}(x) = (x - z) \cdot \nu_E / \rho$ enters.* By contrast, defining the centre field via the Fraenkel functional first-order condition would yield a kinematic formula whose right-hand side involves $V_\rho - H_{z,\rho}$, and the bound would consume the per- ρ homothetic velocity defect (audit-trail markdown note, input (2.4)), itself conditional on the per- ρ radial trace (R)/(1.2), which is open in the no-graph regime. The centroid choice eliminates the homothety because it is defined directly by a bulk integral, not by a minimisation principle comparing to balls. This is precisely the structural mechanism by which Theorem 1.2 avoids (R), (B²), and the per- ρ pointwise radial-trace inputs of the Wave-1 audit (audit-trail markdown note, inputs (1.1), (1.2)=(R), (2.4) in the source notation) entirely.

4 Pointwise bound $|\bar{z}'_\rho|^2 \leq C(D_H + D_I)$

We bound the integrand in (F) pointwise by the two algebraic deficits $D_H(t(\rho)), D_I(t(\rho))$ of [9].

Lemma 4.1 (Pointwise centroid-velocity bound on good levels). *There is $C_* = C_*(n, R, \rho_*)$ such that for a.e. $\rho \in G_{CL}$,*

$$|\bar{z}'_\rho|^2 \leq C_*(D_H(t(\rho)) + D_I(t(\rho)) + \delta_T(\Omega)). \quad (6)$$

Proof. Set $\bar{V}_\rho := P(B_\rho)/P(E_\rho) \in (0, 1]$. Decompose $V_\rho = \bar{V}_\rho + (V_\rho - \bar{V}_\rho)$ in (5):

$$|E_\rho| \bar{z}'_\rho = \underbrace{\bar{V}_\rho \int_{\partial^* E_\rho} (x - \bar{z}_\rho) d\mathcal{H}^{n-1}}_{(I)} + \underbrace{\int_{\partial^* E_\rho} (x - \bar{z}_\rho) (V_\rho - \bar{V}_\rho) d\mathcal{H}^{n-1}}_{(II)}. \quad (7)$$

Bound on (II). By CL, $|x - \bar{z}_\rho| \leq 3$, so $|(II)| \leq 3 \int |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1}$. Set $A_\rho := \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = P(B_\rho)$ (by coarea with $f \equiv 1$ as in the proof of Lemma 3.1); the mean here is the perimeter-mean $\bar{V}_\rho = A_\rho/P(E_\rho)$. The algebraic Cauchy–Schwarz

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq P(E_\rho) \int_{\partial^* E_\rho} (V_\rho - \bar{V}_\rho)^2 d\mathcal{H}^{n-1}$$

together with the variance-form bound $\int (V_\rho - \bar{V}_\rho)^2 d\mathcal{H}^{n-1} \leq C(n, \rho_*) D_H(t(\rho))$ (exact identity of [9], controlling D_H by the L^2 variance of V_ρ around $\bar{V}_\rho$), and the perimeter bound $P(E_\rho) \leq C(n, \rho_*)$ on G_{CL} , give

$$\left(\int |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C_{n, \rho_*} D_H(t(\rho)). \quad (8)$$

Bound on (I). We claim, for each i ,

$$\left| \int_{\partial^* E_\rho} (x - \bar{z}_\rho)_i d\mathcal{H}^{n-1} \right| \leq C_2(n, \rho_*) (\sqrt{\delta_\rho} + \sqrt{\delta_T}), \quad (9)$$

with $\delta_\rho := P(E_\rho) - P(B_\rho)$. To see this, write $(x - \bar{z}_\rho)_i = |x - \bar{z}_\rho| (e_r)_i$, and split $(e_r)_i = (\nu_E)_i + ((e_r)_i - (\nu_E)_i)$.

For the first piece, apply the divergence theorem to the Lipschitz field $V^{(i)}(x) := |x - \bar{z}_\rho| e_i$ (where e_i is the i -th standard basis vector, so $V^{(i)}$ has only its i -th component nonzero). Hence

$$\operatorname{div} V^{(i)}(x) = \partial_i |x - \bar{z}_\rho| = (e_r(x))_i, \quad e_r(x) := \frac{x - \bar{z}_\rho}{|x - \bar{z}_\rho|}.$$

The kink at $x = \bar{z}_\rho$ is a single point; since $n \geq 2$ this set has codimension ≥ 2 , hence is $\mathcal{H}^{n-1}$ -null and contributes no singular part to the distributional gradient. By [7, Thm. 15.9],

$$\int_{\partial^* E_\rho} |x - \bar{z}_\rho| (\nu_E)_i d\mathcal{H}^{n-1} = \int_{E_\rho} (e_r(x))_i dx. \quad (10)$$

On $B_\rho(\bar{z}_\rho)$, $\int (e_r)_i = 0$ by spherical symmetry. Hence $|\int_{E_\rho} (e_r)_i| \leq |E_\rho \Delta B_\rho(\bar{z}_\rho)|$. By FMP [5] at the Fraenkel centre z_ρ^{Fr} , $|E_\rho \Delta B_\rho(z_\rho^{\text{Fr}})| \leq C\sqrt{\delta_\rho}$, while the centroid–Fraenkel proximity $|\bar{z}_\rho - z_\rho^{\text{Fr}}| \leq C'\sqrt{\delta_T}$ (Remark 1.3) costs an additional $|B_\rho(\bar{z}_\rho) \Delta B_\rho(z_\rho^{\text{Fr}})| \leq |\bar{z}_\rho - z_\rho^{\text{Fr}}| \cdot P(B_\rho) \leq C''\sqrt{\delta_T}$ in recentring. Hence

$$|E_\rho \Delta B_\rho(\bar{z}_\rho)| \leq C(\sqrt{\delta_\rho} + \sqrt{\delta_T}).$$

On the good set G_{CL} , both δ_ρ and the recentring cost $\sqrt{\delta_T}$ are controlled, so the rate is preserved.

For the second piece, $|x - \bar{z}_\rho| \leq 3$ and the FJ L^1 oscillation [6] at the centroid (with $O(\sqrt{\delta_\rho})$ change-of-centre error absorbed): $\int |\nu_E - e_r| \leq C_{\text{FJ}}\sqrt{\delta_\rho}$.

Combining gives (9).

Combining. From (7), (8), (9):

$$|E_\rho|^2 |\bar{z}'_\rho|^2 \leq 4C_2^2 \delta_\rho + 4C_2^2 \delta_T + 2C_1 D_H,$$

where $C_1 = 9C_{n, \rho_*}$ (the 9 comes from 3^2 since $|x - \bar{z}_\rho| \leq 3$ on the bounded reduction). Use $\delta_\rho \leq D_I/(n\omega_n \rho_*^{n-1})$ and $|E_\rho|^2 \geq \omega_n^2 \rho_*^{2n}$ to get (6). $\square$

5 Closing the integral

Proof of Theorem 1.2. Split the integral into $G_{\text{CL}} \cup B_{\text{CL}}$. On G_{CL} , Lemma 4.1, (3), and the finiteness $\mu([\rho_*, 1]) < \infty$ give

$$\int_{G_{\text{CL}} \cap [\rho_*, \rho_\delta]} |\bar{z}'_\rho|^2 d\mu(\rho) \leq C_* \int_{\rho_*}^1 (D_H + D_I) d\mu + C_* \delta_T(\Omega) \mu([\rho_*, 1]) \leq C\delta.$$

On B_{CL} , Lemma 2.1 gives $|\bar{z}'_\rho| \leq L_{\text{cent}}$ a.e., while $\mu(B_{\text{CL}}) \leq C\delta$. Hence

$$\int_{B_{\text{CL}} \cap [\rho_*, \rho_\delta]} |\bar{z}'_\rho|^2 d\mu(\rho) \leq L_{\text{cent}}^2 \mu(B_{\text{CL}}) \leq C\delta.$$

Combining the two estimates proves (F). $\square$

6 Conditionality and remarks

Claim: Theorem 1.2 is unconditional. It uses none of: the per- ρ pointwise inputs (1.1), (1.2)=(R), (2.4) of the audit-trail markdown note (open in the no-graph regime); the pointwise quadratic (B²); Agent 3 (7.1); any selection, graph entry, BDV minimiser, Schauder, or Fuglede expansion.

Structural reason: the centroid kinematic identity (5) has no homothety term, so the bound goes through the unconditional pair (Cauchy–Schwarz on V_ρ ; FMP+FJ+divergence at the centroid).

Remark 6.1 (Sharpness). The pointwise bound (6) is sharp in scaling: for a small translation of a ball, $V_\rho - \bar{V}_\rho$ has dipole structure of size $|\bar{z}'_\rho|$, giving $D_H \sim |\bar{z}'_\rho|^2$.

Remark 6.2 (Outside the smallness regime). If $\delta_T(\Omega) > \delta_0$, then $\mathcal{A}(\Omega) \geq c(n) > 0$ by FMP, and (F) is trivial after enlarging C_{cent} .

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